Implicit Field Supervision for Robust Non-Rigid Shape Matching

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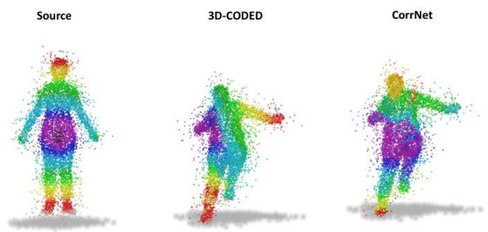
Motivation

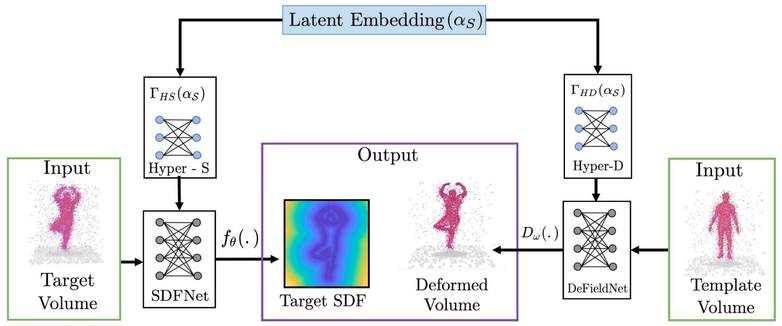
- Existing non-rigid shape matching algorithms show weak resilience in the presence of noise, outliers, etc.
- To overcome, we leverage the expressive power of deep implicit functions.



Method Overview

- For input $\boldsymbol{X}$, we learn a deformation field through DeFieldNet to deform template volume $\boldsymbol{T}$ to match $\mathbf{X}$.
- Simultaneously, we learn the zero-level set of $\boldsymbol{X}$ through SDFNet.
- Shape-specific $\boldsymbol{\alpha}_S$ are optimised by surface-level supervision and volumetric regularization.

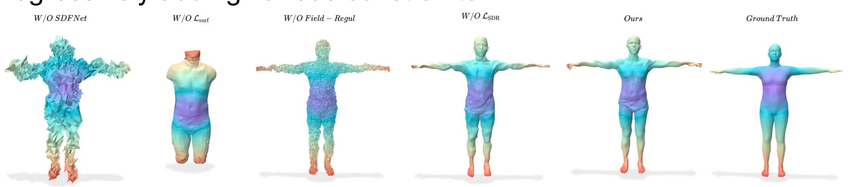
$$\mathcal{E}_{\text{Train}} = \Lambda_1 \mathcal{L}_{\mathcal{SDF}} + \Lambda_2 \mathcal{L}_{surf} + \Lambda_3 \mathcal{L}_{\mathcal{SDR}} + \Lambda_4 \mathcal{L}_{Smooth} + \Lambda_5 \mathcal{L}_{vol}$$



- L_{SDF} is used to train SDFNet with normal and SDF supervision terms, alike SIREN[5]
- L_{Surf} supervises corresponding points on the surface, L_{smooth} and L_{vol}, L_{SDR} are regularisations.

Ablating constraints

Progressively adding various constraints



Signed Distance Regularization (SDR)

Points close to surface undergo minimal change in SDF under deformation field.
While we know σ_t, the GT SDF of point t, its SDF after deformation σ_x, is measured using RBF interpolation

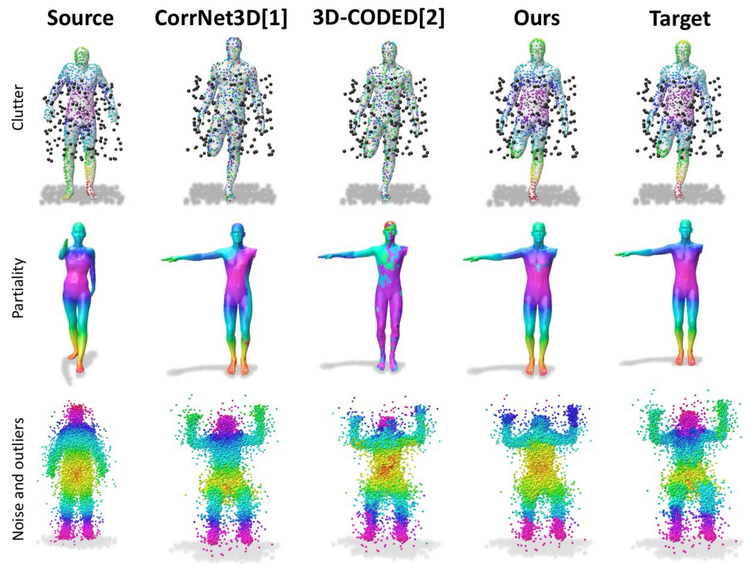
$$\hat{\sigma}_{\hat{x}_i} = \varphi(\hat{x}_i)\, \Phi^{-1} \Delta$$

The net loss, is then computed as

$$\mathcal{L}_{SDR} = \sum_{t_i \in [\mathcal{T}]} ||\,\text{clamp}(\sigma_{t_i}, \eta) - \text{clamp}(\hat{\sigma}_{\hat{x}_i}, \eta)||_2$$

Results on compromised data (Synthetic)

We introduce three main type of corruptions namely, noise, outliers and clutter points to shapes of FAUST and SHREC19 dataset.



Method	Dif-FMap[3]	GeoFMap[4]	CorrNet3D[2]	3D-CODED[1]	Ours
Clutter	17.1	52.2	51.1	22.1	8.1
Outlier	16.7	27.8	36.0	14.4	7.7
Partiality	26.3	48.6	23.5	6.0	4.3

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Inference

We follow three staged inference to extract a point-wise map $\boldsymbol{\pi}$ between pair of shapes $\boldsymbol{X}$ and $\boldsymbol{Y}$,

- We first solve for $\boldsymbol{\alpha}_i$ (and network weights) given a pointset w/ or w/o SDF values by MAP estimate

$$\alpha_i = \underset{\alpha_i}{\text{argmin}} \;\; \Lambda_1 \mathcal{L}_{\mathcal{SDF}} + \Lambda_3 \mathcal{L}_{\mathcal{SDR}}$$

$$\omega := \Gamma_{HD}(\alpha_i)$$

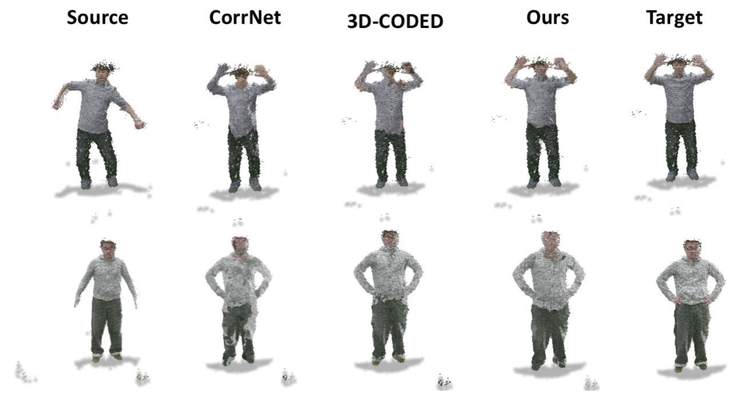
- We further enhance reconstruction by minimising Chamfer's Distance

$$\alpha_{\mathbf{opt}} = \underset{\alpha_i}{\text{argmin}} \sum_{\tilde{\mathbf{s}} \in \partial \mathcal{S}} \min_{\tilde{\mathbf{t}}_i \in \partial \mathcal{T}} \left|D_\omega(\tilde{\mathbf{t}}_i) - \tilde{\mathbf{s}}\right|^2 + \sum_{\tilde{t}_i \in \partial \mathcal{T}} \min_{\tilde{\mathbf{s}} \in \partial \mathcal{S}} \left|D_\omega(\tilde{\mathbf{t}}_i) - \tilde{\mathbf{s}}\right|^2$$

- Correspondence is then computed using NN-search

Results on Real Data

We compare our approach with baselines on the CMU-Panoptic dataset.



Method	Dif-FMap[3]	GeoFMap [4]	3D-CODED[2]	CorrNet3D[1]	Ours
Keypoint Error	39.4	23.9	17.1	14.8	**8.5**

Conclusion

- Demonstrate the use of Neural Fields in successfully establishing correspondence between non-rigid objects.
- The resulting method shows strong resilience towards artefacts.

References

[1] 3D-CODED : 3D Correspondences by Deep Deformation, T. Groueix et.al, ECCV 2018
[2] CorrNet3D: Unsupervised End-to-end Learning of Dense Correspondence for 3D Point Clouds, Zeng et.al, CVPR 2021
[3]Correspondence Learning via Linearly-invariant Embedding, Marin et.al, NeurIPS 2020,
[4] Deep geometric functional maps, Donati et.al, CVPR 2020
[5] Implicit Neural Representations with Periodic Activation Functions. Sitzmann et.al, NeurIPS 2020